# Why Bitter Coffee Starts to Taste Better

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Your first sip of black coffee can taste awful. Its sharp bitterness is the kind of flavor our brains are wired to avoid. Yet millions of people grow to love that same taste. What changes? Scientists at UC San Francisco may have found an important part of the answer.

The team studied **dopamine**, a chemical messenger involved in motivation, reward and learning. Most research has looked at fast **dopamine** signals that fire within seconds of something rewarding. The new study, published November 4 in Nature, looks at a slower system instead. It responds to information from inside the body, helping the brain learn about the benefits of food after it has been eaten.

When food reaches the gut, sensors detect **nutrients** such as sugar and fat. That information travels to the brain, where it can trigger a **dopamine** response that builds slowly and lasts several minutes. Senior author Zachary Knight and graduate student James Grove gave mice sugar water or a milkshake-like liquid straight into their stomachs, and watched **dopamine** activity across their brains.

The strongest response appeared in the anterior basolateral **amygdala**, or BLA, a brain region involved in emotion and learning. "We've shown that this slow **dopamine** system is what drives us to learn to like certain foods over time, and we can see where and how that happens," Knight said.

The team then tested whether the signal really changes food preferences. When they interrupted the **pathway**, mice still ate normally but could no longer learn that an unfamiliar flavor meant calories were coming. When they switched those fibers on after mice tasted a calorie-free flavor, the animals still developed a liking for it. Once the brain links coffee's taste with **caffeine**'s stimulating effects, Knight explained, "that coffee starts to taste a lot better."

## Word Treasure

**dopamine** -- A chemical messenger in the brain that helps with reward and learning.

**amygdala** -- A part of the brain that helps with feelings and learning.

**nutrients** -- Useful parts of food, such as sugar and fat, that the body needs.

**caffeine** -- The substance in coffee that wakes you up and makes you feel alert.

**pathway** -- A route that signals travel along in the body or brain.

## Background you need

Dopamine is a chemical messenger in the brain that helps us feel reward and learn from what happens to us.

The gut is the part of the body where food is digested, and it sends signals to the brain about what we have eaten.

Learning to like a food takes time: the brain links a flavour with how the food makes the body feel after it is eaten.

Coffee contains caffeine, a substance that makes people feel more awake, which is one reason adults often drink it.

## Break it down

**LEAD:** Scientists at UC San Francisco may have found why bitter coffee starts to taste better.

**KEY EVENT:** A study published November 4 in Nature describes a slow dopamine system from the gut to the brain.

+- WHO: Senior author Zachary Knight and graduate student James Grove.

+- WHAT: A slow dopamine response that builds over minutes and teaches the brain to like foods.

+- WHERE: The strongest response appeared in the anterior basolateral amygdala (BLA).

**EVIDENCE:** Blocking the pathway stopped mice learning a flavour meant calories, and switching the fibres on made them like a calorie-free flavour.

**TAKEAWAY:** Once the brain links coffee's taste with caffeine's stimulating effects, coffee starts to taste a lot better.

## Why it matters

This helps explain how our likes and dislikes are learned, not just born with us, and why foods that taste strange at first can become favourites over time.

## Test what you remember

**1. What is dopamine?**

- (A) A chemical messenger involved in motivation, reward and learning
- (B) A nutrient found in coffee
- (C) A type of sugar
- (D) A part of the stomach

**2. What does the slower dopamine system respond to?**

- (A) The colour of food
- (B) Sounds made while eating
- (C) Information from inside the body
- (D) The smell of food

**3. What do sensors in the gut detect when food arrives?**

- (A) The taste of salt
- (B) Nutrients such as sugar and fat
- (C) The size of each bite
- (D) The temperature of the food

**4. In which brain region did the strongest dopamine response appear?**

- (A) The anterior basolateral amygdala (BLA)
- (B) The part of the brain that controls hearing
- (C) The part of the brain that controls balance
- (D) The part of the brain that controls sight

**5. What happened when the team interrupted the pathway in mice?**

- (A) The mice stopped eating
- (B) The mice ate much more than before
- (C) The mice still ate normally but could no longer learn that an unfamiliar flavour meant calories were coming
- (D) The mice refused all food

**6. When was the study published in Nature?**

- (A) November 7
- (B) November 4
- (C) November 14
- (D) October 4

Answer key (try the quiz first!): 1: A 2: C 3: B 4: A 5: C 6: B

# Questions to think about

## MY THOUGHTS (at least 20 words):